

RAND Doctor-Visits Expanded Workflow Analysis

Annotated native gamlss.longitudinal workflow replication with minimal code for each output

Executive Summary

The RAND Health and Retirement Study doctor-visits analysis used a deterministic expanded-profile sample of 200 subjects from 857 eligible complete-covariate subjects (2726 observed responses). The primary model selected by joint AIC used a NBI marginal distribution with a Frank copula and theta formula $\sim$ hospital_stay (joint AIC 14715.33). The pre-specified covariate-dependence contrast $\sim$ wave_scaled + arthritis was a likelihood-improving exploratory contrast rather than the AIC-preferred model: joint log likelihood changed by 1.95, AIC changed by 0.09, and the nested LR-screen p-value was 0.142. The primary refit reported convergence as TRUE after 3 outer iterations, using convergence fields read directly from the fitted object. Following the native workflow, the analysis also records data shape, missingness, time-stratified moments, pairwise dependence, screening overlays, check_model(), term plots, marginal diagnostics, copula diagnostics, predictions, marginal effects, model-based simulations, and benchmark-standard-model comparisons. The compact check_model() statuses were: Convergence=PASS; Marginal fit=FAIL; Tail fit=PASS; Copula fit=PASS; Variance calculation=PASS. The associated figures report the observed doctor-visit distribution, adjacent-wave Kendall dependence, fitted covariate effects on the selected copula dependence parameter, and available marginal-parameter standard errors. Exact-discrete longitudinal standard errors are marked unavailable when numerical Hessian estimation is singular or intentionally skipped.

Quantity	Value
Profile	expanded
Sample	200 subjects; 2726 observed responses
Eligible subjects	857.00
Selected margin	NBI
Selected copula	Frank
Primary theta model	$\sim$ hospital_stay (AIC 14715.3)
Convergence	TRUE after 3 outer iterations

Workflow Map

The report follows the native-simulation-workflow.html sequence, adapted for an observed count outcome: inspect data shape and missingness; review marginal moments and pairwise dependence; screen marginal family; screen copula family; fit theta covariate candidates; review the final model with check_model, term plots, marginal diagnostics, and copula diagnostics; then answer applied questions using benchmarks, prediction, marginal effects, and simulation.

```
settings <- jss_settings()
data <- jss_rand_load_data(settings)
prepared <- jss_rand_prepare_data(data)
sample_data <- jss_rand_sample_data(prepared$data, settings)
fit_budget <- jss_rand_fit_budget(settings)
```

Workflow Artifact Status

This audit table is a reproducibility guardrail. Each workflow step records whether its table or plot was created, so later manuscript builds can distinguish substantive model diagnostics from failed rendering or missing optional dependencies.

workflow_step	status	exists	bytes	error
data_shape	ok	TRUE	335.00	
missingness_check	ok	TRUE	65.00	
time_moments	ok	TRUE	1384.0	
pairwise_correlation	ok	TRUE	4406.0	
check_model	ok	TRUE	711.00	
Response and Pairwise Dependence Matrix	ok	TRUE	1801304.0	
Final Margin Overlay	ok	TRUE	49899.0	
Final Margin Overlay by Wave	ok	TRUE	232758.0	
Screened Copula Fit by Adjacent Wave	ok	TRUE	917966.0	
Final Model Term Plots	ok	TRUE	253932.0	
Marginal Diagnostics	ok	TRUE	266677.0	
Copula Diagnostics	ok	TRUE	1392113.0	
benchmark_standard_models	ok	TRUE	2820.0	
prediction	ok	TRUE	1629.0	
marginal_effects	ok	TRUE	202.00	
simulate	ok	TRUE	1030.0	

Step 1: Data Shape Check

The native workflow begins by verifying that the data are a long panel with one subject-time record per row, a response column, a subject identifier, and an ordered time variable. For RAND, the time variable is the numeric HRS/RAND wave and the response is doctor visits.

```
shape <- jss_rand_data_shape_table(full_data, sample_data, settings)
utils::write.csv(shape, paths$data_shape, row.names = FALSE)
```

profile	full_rows	full_subjects	full_waves	sample_rows	sample_subjects	sample_waves	response_column	subject_column	time_column
expanded	593670.0	42405.0	14.00	2800.0	200.00	14.00	doctor_visits	subject	time

Step 2: Missingness Check

The workflow next asks whether response missingness is associated with observed predictors. This does not prove MCAR or rule out MNAR, but it records whether the observed covariates explain missingness patterns that should be acknowledged in the application.

```
missingness_check <- check_missingness(
  full_data,
  response_var = "doctor_visits",
  time_var = "time",
  subject_var = "subject",
  predictors = c("time", "hospital_stay", "self_rated_health", "hypertension",
    "diabetes", "cancer", "heart_problem", "stroke",
    "arthritis", "sex", "education", "age_scaled")
)
```

n	n_missing	prop_missing
593670.0	339728.0	0.572

Missingness Predictor Screen

Predictors with usable variation enter the missingness model. This mirrors the vignette check and documents which covariates were eligible for the missingness screen.

predictor	class	n_observed	n_unique	used	reason
time	integer	593670.0	14.00	TRUE	used
hospital_stay	factor	266385.0	2.000	TRUE	used
self_rated_health	factor	267491.0	5.000	TRUE	used
hypertension	factor	260553.0	2.000	TRUE	used
diabetes	factor	264742.0	2.000	TRUE	used
cancer	factor	266265.0	2.000	TRUE	used
heart_problem	factor	265215.0	2.000	TRUE	used
stroke	factor	265312.0	2.000	TRUE	used
arthritis	factor	260487.0	2.000	TRUE	used
sex	factor	593670.0	2.000	TRUE	used
education	factor	593362.0	5.000	TRUE	used
age_scaled	numeric	593656.0	130.00	TRUE	used

Missingness Term Tests

When the missingness logistic regression is estimable, term-level tests are recorded. These are screening diagnostics rather than causal missingness tests.

term	statistic	df	p_value
time	3184.4	1.000	0.00e+00
hospital_stay	838.94	1.000	1.85e-184
self_rated_health	1392.2	4.000	3.43e-300
hypertension	76.47	1.000	2.24e-18
diabetes	68.90	1.000	1.04e-16
cancer	40.69	1.000	1.78e-10

term	statistic	df	p_value
heart_problem	2.318	1.000	0.128
stroke	26.64	1.000	2.45e-07
arthritis	27.81	1.000	1.34e-07
sex	296.61	1.000	1.81e-66
education	447.33	4.000	1.64e-95
age_scaled	823.06	1.000	5.24e-181

Step 3: Time-Stratified Response Moments

Before fitting, the workflow inspects whether mean, variance, zero mass, skewness, and kurtosis change over wave. For the doctor-visit count outcome this helps motivate a flexible count family and time/covariate adjustment in the marginal model.

```
time_moments <- aggregate_by_wave(sample_data, doctor_visits)
# mean, variance, skewness, kurtosis, and zero percentage by RAND wave
```

wave	survey_year	n	mean	variance	skewness	kurtosis	zero_pct
2.000	1994.0	198.00	3.702	28.24	6.533	65.29	13.64
3.000	1996.0	200.00	4.680	43.64	4.309	26.31	11.50
4.000	1998.0	199.00	5.427	67.38	4.869	33.99	10.05
5.000	2000.0	200.00	5.600	40.83	3.752	28.18	10.00
6.000	2002.0	198.00	5.808	38.66	3.039	17.25	9.091
7.000	2004.0	199.00	5.910	41.95	3.343	20.70	8.040
8.000	2006.0	199.00	7.060	76.88	6.718	66.39	5.025
9.000	2008.0	196.00	7.143	73.77	3.481	17.97	5.612
10.00	2010.0	198.00	8.333	98.71	5.790	53.63	6.061
11.00	2012.0	196.00	6.235	30.43	1.439	4.894	9.694
12.00	2014.0	193.00	8.036	70.49	2.881	15.17	8.808
13.00	2016.0	185.00	7.914	68.78	3.056	17.37	5.946
14.00	2018.0	187.00	8.011	64.81	2.205	9.450	8.556
15.00	2020.0	178.00	8.579	76.70	2.065	8.850	10.67

Step 4: Pairwise Dependence Inspection

The pairwise correlation table is the tabular complement to plot_dist. It shows how association changes with wave pair and lag before any copula is fitted.

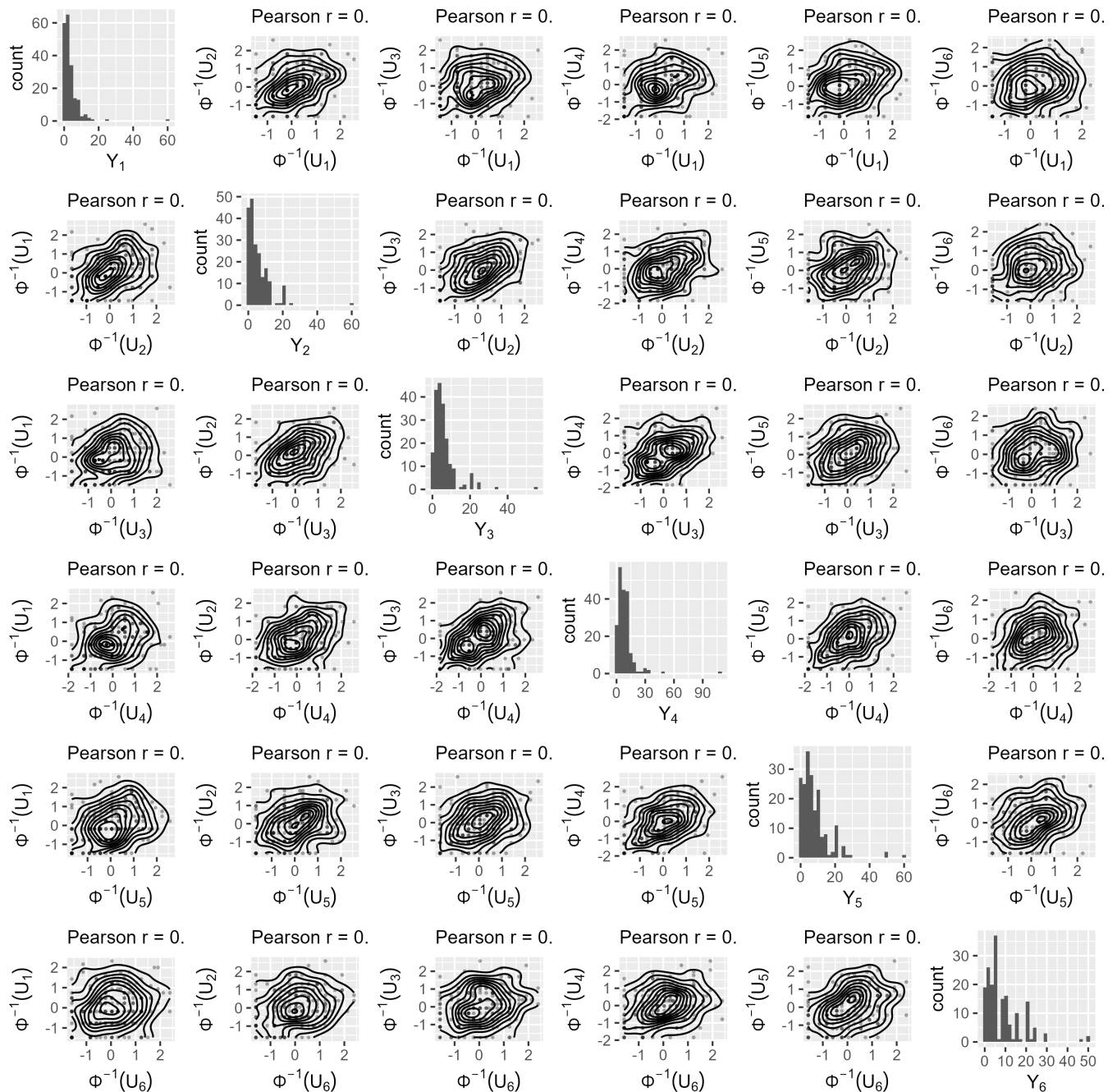
```
wide <- reshape(sample_data[c("subject", "wave", "doctor_visits")],
idvar = "subject", timevar = "wave", direction = "wide")
cor(wide[-1], use = "pairwise.complete.obs", method = "kendall")
```

wave_left	wave_right	lag	n_pairs	pearson	kendall
2.000	3.000	1.000	198.00	0.319	0.391
2.000	4.000	2.000	197.00	0.324	0.346
2.000	5.000	3.000	198.00	0.258	0.321
2.000	6.000	4.000	196.00	0.503	0.303
2.000	7.000	5.000	197.00	0.183	0.213
2.000	8.000	6.000	197.00	0.327	0.240
2.000	9.000	7.000	194.00	0.159	0.271
2.000	10.00	8.000	196.00	0.183	0.278
2.000	11.00	9.000	194.00	0.165	0.256
2.000	12.00	10.00	191.00	0.100	0.268
2.000	13.00	11.00	183.00	0.209	0.260
2.000	14.00	12.00	185.00	0.163	0.198
2.000	15.00	13.00	176.00	0.138	0.158
3.000	4.000	1.000	199.00	0.496	0.475
3.000	5.000	2.000	200.00	0.351	0.374
3.000	6.000	3.000	198.00	0.290	0.242
3.000	7.000	4.000	199.00	0.148	0.262
3.000	8.000	5.000	199.00	0.269	0.369

Step 4 Plot: Response and Pairwise Dependence Matrix

This is the native `plot_dist` diagnostic adapted to the count response. The diagonal panels show the marginal doctor-visit distributions and the off-diagonal panels show normal-score pseudo-observation dependence. Representative waves are used to keep the grid readable, while the full pairwise table covers every wave pair.

```
plot_dist(
  representative_wave_data,
  margin_dist = NBI(),
  subject_var = "subject",
  time_var = "time",
  response_var = "doctor_visits",
  offdiag_scale = "pseudo",
  transform = "normal"
)
```



Representative waves only; all waves are covered in the pairwise-correlation table.

Step 5: Marginal Count Family Screen

The vignette uses `select_margin` for continuous positive data. Here the equivalent step screens count families that are plausible for doctor visits. NBI is retained because it improves AIC/BIC relative to Poisson and zero-inflated alternatives in this sampled application.

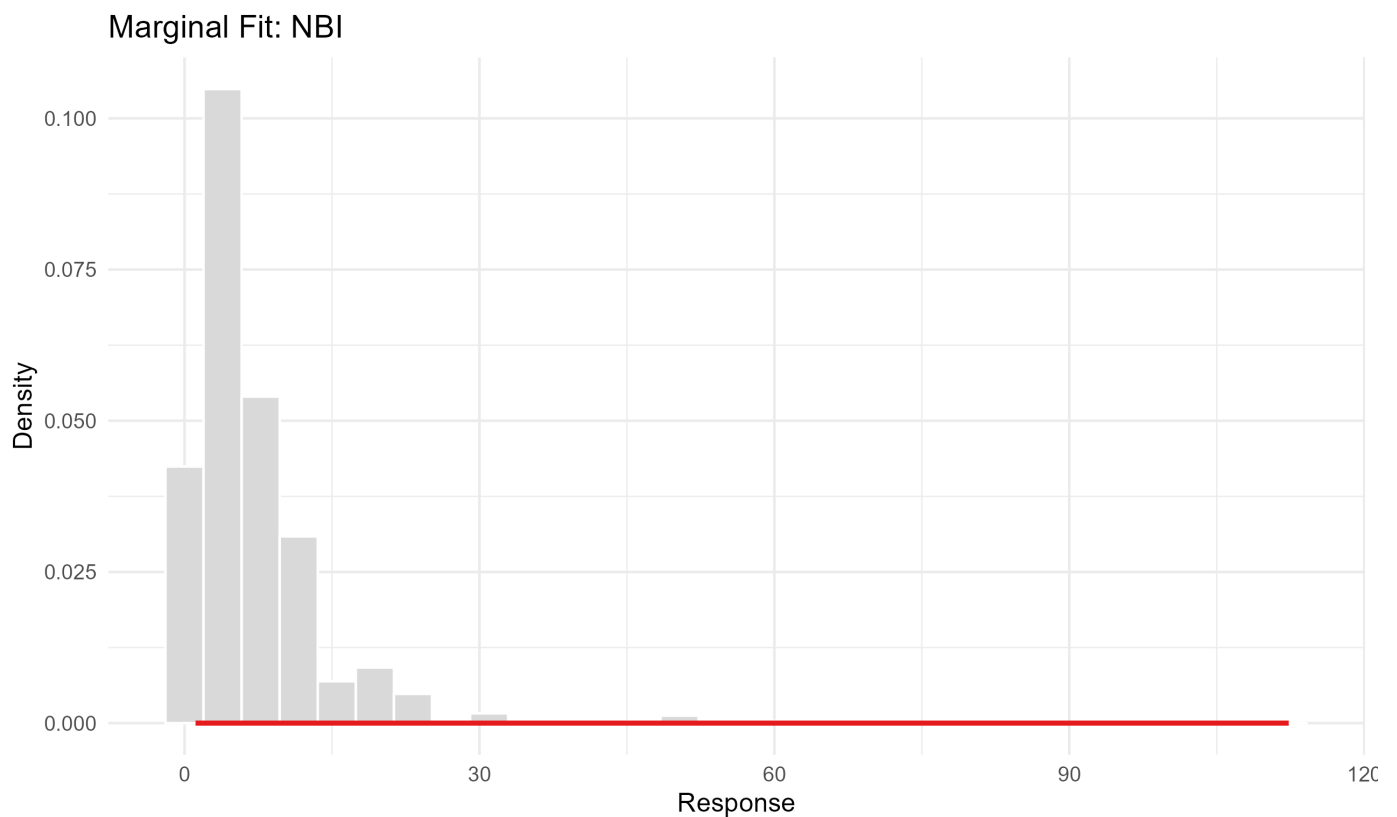
```
family_screen <- jss_rand_family_screen(sample_data)
selected_family <- jss_rand_select_margin_family(family_screen$table)
```

family	aic	bic	converged	error
NBI	15350.1	15539.3	TRUE	
NBII	15369.4	15558.6	TRUE	
ZIP	21041.2	21230.4	TRUE	
PO	22007.1	22154.8	TRUE	

Step 5 Plot: Final Marginal Fit Overlay

After selecting the margin and fitting covariates, this plot checks whether the fitted NBI marginal distribution follows the observed response distribution overall.

```
plot_margin_fit(final_fit, bins = 30, plot = FALSE)$plot
```

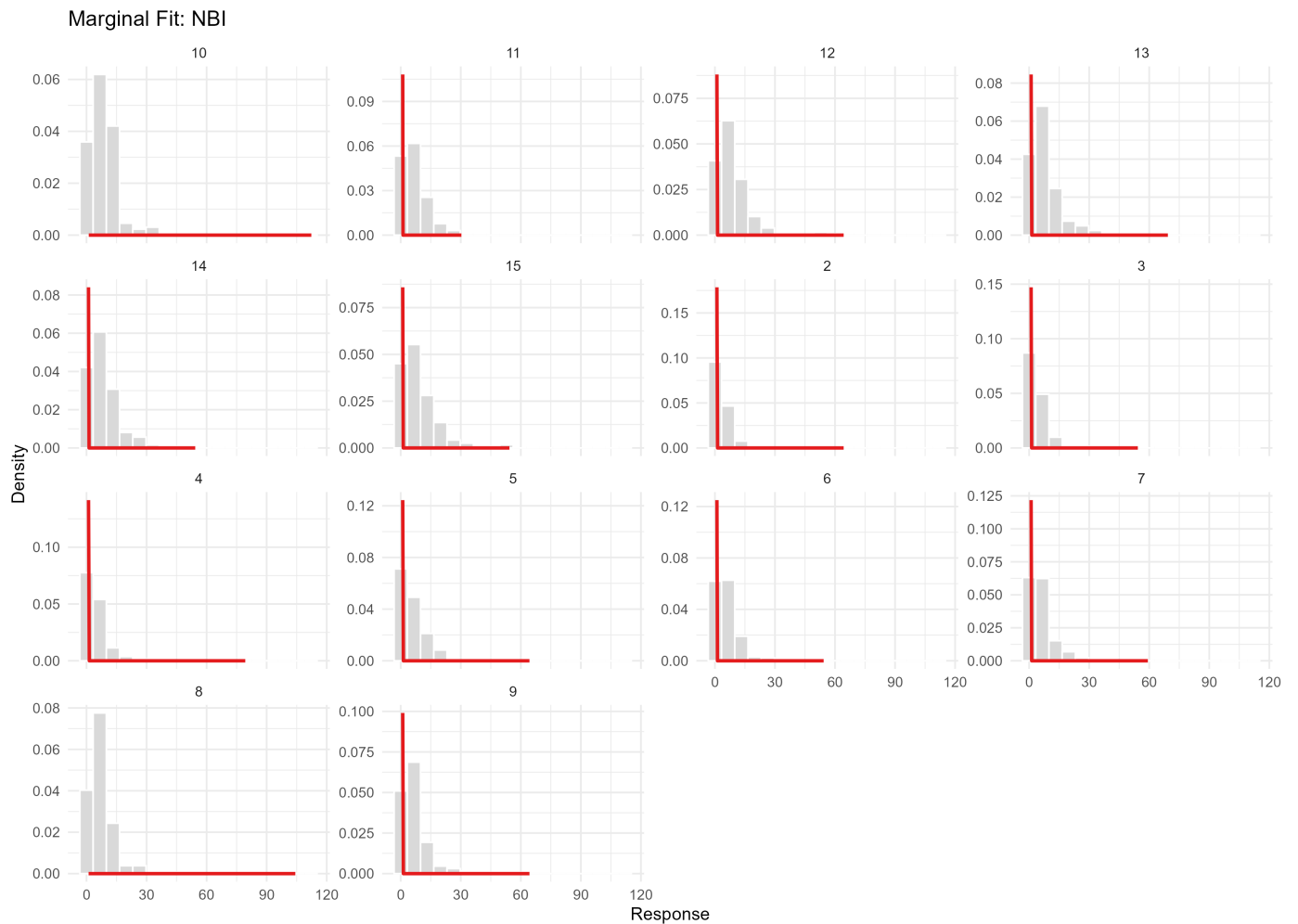


Overall marginal overlay for the final fitted model.

Step 5 Plot: Final Marginal Fit by Wave

The native workflow then repeats the marginal overlay by time. This exposes wave-specific lack of fit that an aggregate histogram can hide.

```
plot_margin_fit(final_fit, bins = 18, by_time = TRUE, plot = FALSE)$plot
```



Wave-stratified marginal overlay for the final fitted model.

Step 6: Copula Family Screen

With the marginal family fixed, the workflow screens copula families on the joint likelihood. In the expanded RAND sample the Frank copula is selected ahead of Gaussian and Clayton.

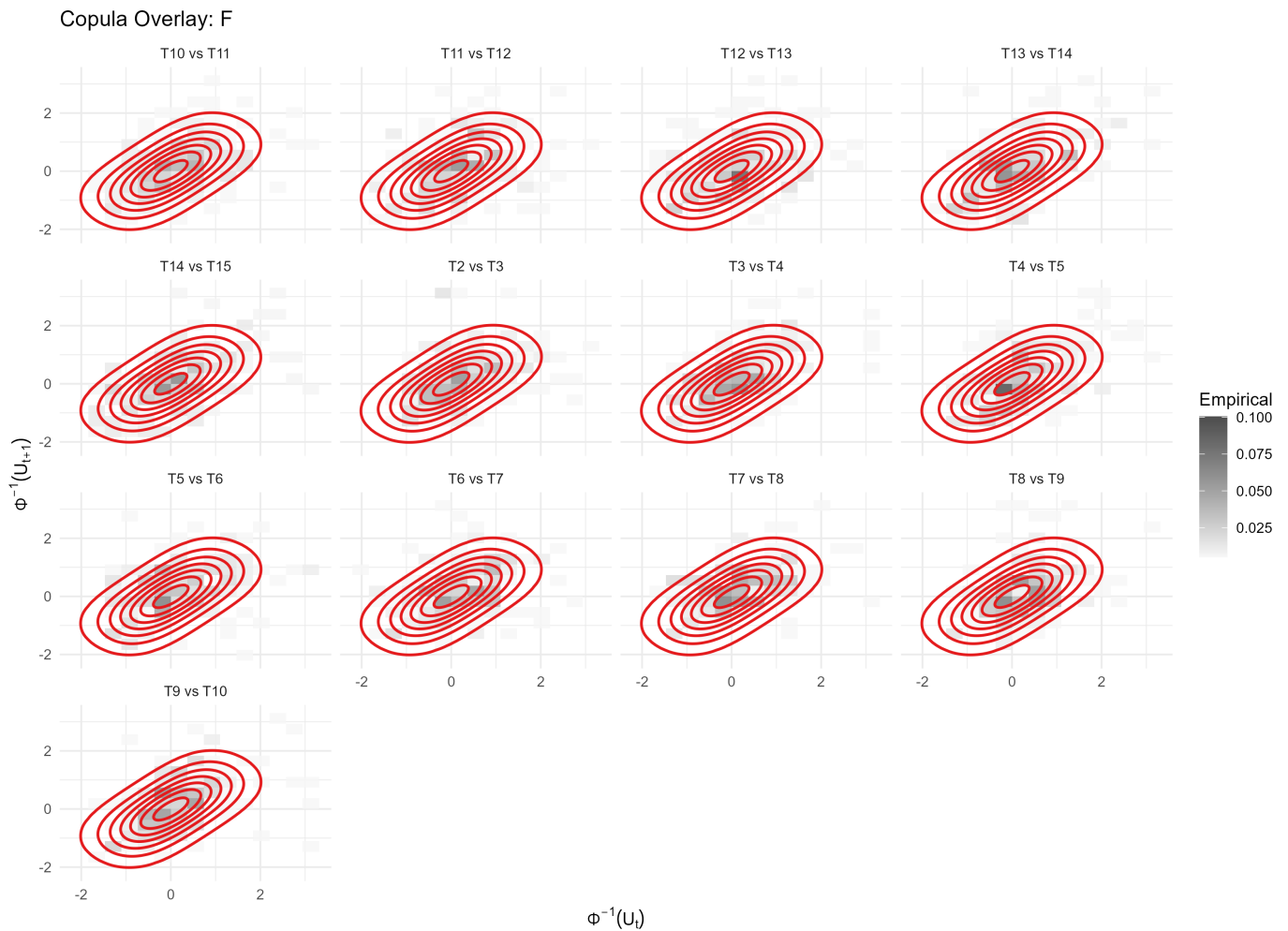
```
copula_screen <- jss_rand_copula_screen(  
  sample_data, fit_budget, settings$profile, selected_family  
)  
selected_copula <- jss_rand_select_copula(copula_screen)
```

copula_name	theta_formula	aic_joint	bic_joint	converged	outer_iterations
Frank	~1	14719.4	14915.4	TRUE	3.000
Gaussian	~1	14807.0	15002.9	TRUE	5.000
Clayton	~1	14808.0	15003.9	TRUE	4.000

Step 6 Plot: Screened Copula Fit

The copula overlay checks whether the selected copula can reproduce the empirical pseudo-observation dependence by adjacent wave. This is the count-data analogue of the vignette `plot_copula_fit` screen.

```
plot_copula_fit(fit = final_fit, by_time = TRUE, bins = 14, plot = FALSE)$plot
```



Empirical adjacent-wave pseudo-observations with fitted copula overlay.

Step 7: Theta Covariate Screen

The main scientific question is whether dependence is covariate-sensitive. This screen compares an intercept-only theta model with wave, arthritis, hospital stay, and broader health covariate candidates. The AIC-preferred expanded model is $\text{theta} \sim \text{hospital_stay}$; the pre-specified wave-plus-arthritis contrast improves likelihood but is not AIC-preferred.

```
theta_screen <- jss_rand_theta_screen(
  sample_data, fit_budget, settings$profile, selected_family, selected_copula
)
theta_screen$stable <- jss_rand_add_theta_evidence(theta_screen$stable)
primary <- jss_rand_select_primary_fit(theta_screen$fits, theta_screen$stable)
contrast <- jss_rand_select_covariate_contrast(theta_screen$fits, theta_screen$stable)
```

model	theta_formula	aic_joint	delta_aic_vs_intercept	lr_p_value_vs_intercept	evidence_label
theta_hospital_stay	~hospital_stay	14715.3	-4.092	0.014	aic_support
theta_stroke	~stroke	14715.6	-3.800	0.016	aic_support
theta_arthritis	~arthritis	14717.7	-1.689	0.055	likelihood_gain_only
theta_intercept	~1	14719.4	0.00e+00		reference
theta_wave_arthritis	~wave_scaled + arthritis	14719.5	0.092	0.142	likelihood_gain_only
theta_heart_problem	~heart_problem	14719.8	0.415	0.208	likelihood_gain_only
theta_diabetes	~diabetes	14721.1	1.702	0.585	likelihood_gain_only
theta_wave	~wave_scaled	14721.3	1.869	0.718	likelihood_gain_only
theta_age	~age_scaled	14721.4	1.954	0.830	likelihood_gain_only
theta_selfRatedHealth	~selfRatedHealth	14723.0	3.542	0.348	likelihood_gain_only

Step 8: Convergence Audit

The archived final fit was not reused as evidence. The expanded module refits with a higher iteration budget and records convergence fields directly from each fitted object.

```
convergence <- jss_rand_convergence_table(
  theta_fits = theta_screen$fits,
  selected_model = primary$model
)
```

model	selected	converged	hit_outer_limit	outer_iterations	max_outer_iter	stop_reason
theta_intercept	FALSE	TRUE	FALSE	3.000	60.00	tolerance
theta_wave	FALSE	TRUE	FALSE	3.000	60.00	tolerance
theta_arthritis	FALSE	TRUE	FALSE	4.000	60.00	tolerance
theta_wave_arthritis	FALSE	TRUE	FALSE	4.000	60.00	tolerance
theta_diabetes	FALSE	TRUE	FALSE	3.000	60.00	tolerance
theta_hospital_stay	TRUE	TRUE	FALSE	3.000	60.00	tolerance
theta_stroke	FALSE	TRUE	FALSE	3.000	60.00	tolerance
theta_age	FALSE	TRUE	FALSE	3.000	60.00	tolerance
theta_selfRatedHealth	FALSE	TRUE	FALSE	4.000	60.00	tolerance
theta_heart_problem	FALSE	TRUE	FALSE	3.000	60.00	tolerance

Step 8: Final Coefficients

This table is the numerical summary of the selected joint model. It should be read together with the dependence diagnostics, because dependence modelling changes uncertainty and the interpretation of marginal terms.

```
coefficient_table <- jss_rand_coefficient_table(final_fit, primary$model)
```

parameter	term	estimate	selected_theta_model
mu	mu.intercept	0.467	theta_hospital_stay
mu	mu.wave_factor3	0.211	theta_hospital_stay
mu	mu.wave_factor4	0.208	theta_hospital_stay
mu	mu.wave_factor5	0.323	theta_hospital_stay
mu	mu.wave_factor6	0.303	theta_hospital_stay
mu	mu.wave_factor7	0.290	theta_hospital_stay
mu	mu.wave_factor8	0.360	theta_hospital_stay
mu	mu.wave_factor9	0.405	theta_hospital_stay
mu	mu.wave_factor10	0.466	theta_hospital_stay
mu	mu.wave_factor11	0.241	theta_hospital_stay
mu	mu.wave_factor12	0.408	theta_hospital_stay
mu	mu.wave_factor13	0.413	theta_hospital_stay
mu	mu.wave_factor14	0.381	theta_hospital_stay
mu	mu.wave_factor15	0.296	theta_hospital_stay
mu	mu.self_rated_healthVery good	0.245	theta_hospital_stay
mu	mu.self_rated_healthGood	0.420	theta_hospital_stay
mu	mu.self_rated_healthFair	0.801	theta_hospital_stay
mu	mu.self_rated_healthPoor	1.381	theta_hospital_stay
mu	mu.hospital_stayYes	0.413	theta_hospital_stay
mu	mu.hypertensionYes	0.244	theta_hospital_stay
mu	mu.cancerYes	0.271	theta_hospital_stay
mu	mu.educationGED	0.460	theta_hospital_stay
mu	mu.educationHigh school graduate	0.515	theta_hospital_stay
mu	mu.educationSome college	0.678	theta_hospital_stay

Step 9: check_model Basic Diagnostics

The native workflow uses `check_model` as the compact numerical diagnostic gate. For the expanded RAND fit, convergence, tail fit, copula residual dependence, and variance calculation pass, but the marginal PIT KS check fails. This is important: the dependence result is usable as a methodological demonstration, but the marginal count model still has calibration imperfections.

```
check <- check_model(final_fit, include_vcov = FALSE, include_plots = FALSE)
check$basic_checks
```

area	status	value	threshold_condition	message
Convergence	PASS	TRUE	Not TRUE	Convergence was confirmed.
Marginal fit	FAIL	0.000000000000000000000 0000002363	ks_p_value < 0.05	The marginal distribution is off by the PIT uniformity screen.
Tail fit	PASS	1.049	max(lower_ratio, upper_ratio) > 2	The basic PIT tail-ratio screen did not flag tail misfit.
Copula fit	PASS	0.190	abs(lag1_cor) > dependence_cor_cutoff	The lag-1 Rosenblatt residual correlation screen did not flag residual dependence.
Variance calculation	PASS	analytical	vcov_method == "numderiv"	The variance-covariance method did not trigger the numerical-Hessian review screen.

check_model Scores and Tail Calibration

The scoring and tail tables provide the components behind the compact check. The PIT tail ratios pass even though the overall PIT uniformity test fails.

```
check$scores
check$tail
```

logs	mae	mse	dss
2.807	4.236	52.43	127.67

PIT Tail Summary

threshold	lower_observed	lower_expected	lower_ratio	upper_observed	upper_expected	upper_ratio
0.050			0.088			1.049
0.100			0.290			0.950

Residual Dependence Summary

This table is the numerical copula check used by `check_model`. Low residual lag correlation after the fitted copula supports the claim that explicit dependence modelling is capturing longitudinal association.

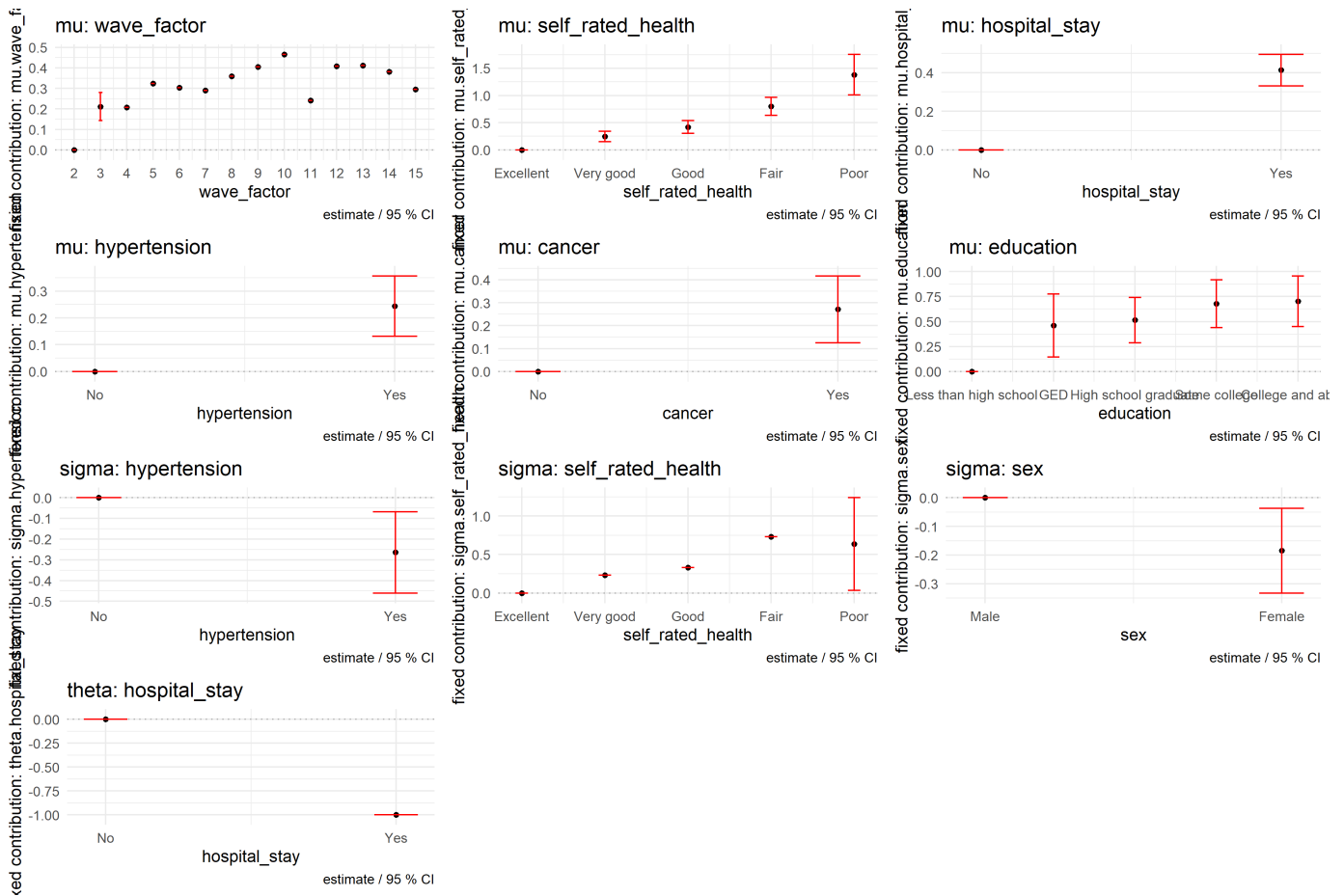
```
check$residual_dependence
```

lag	normal_score_cor	cutoff
1.000	-0.190	0.250

Step 10 Plot: Term Effects

The term plot dashboard is the visual coefficient review from the native workflow. It shows estimated fixed effects and uncertainty for the fitted marginal and theta parameters.

```
plot_terms(final_fit, data = sample_data, ncol = 3)
```

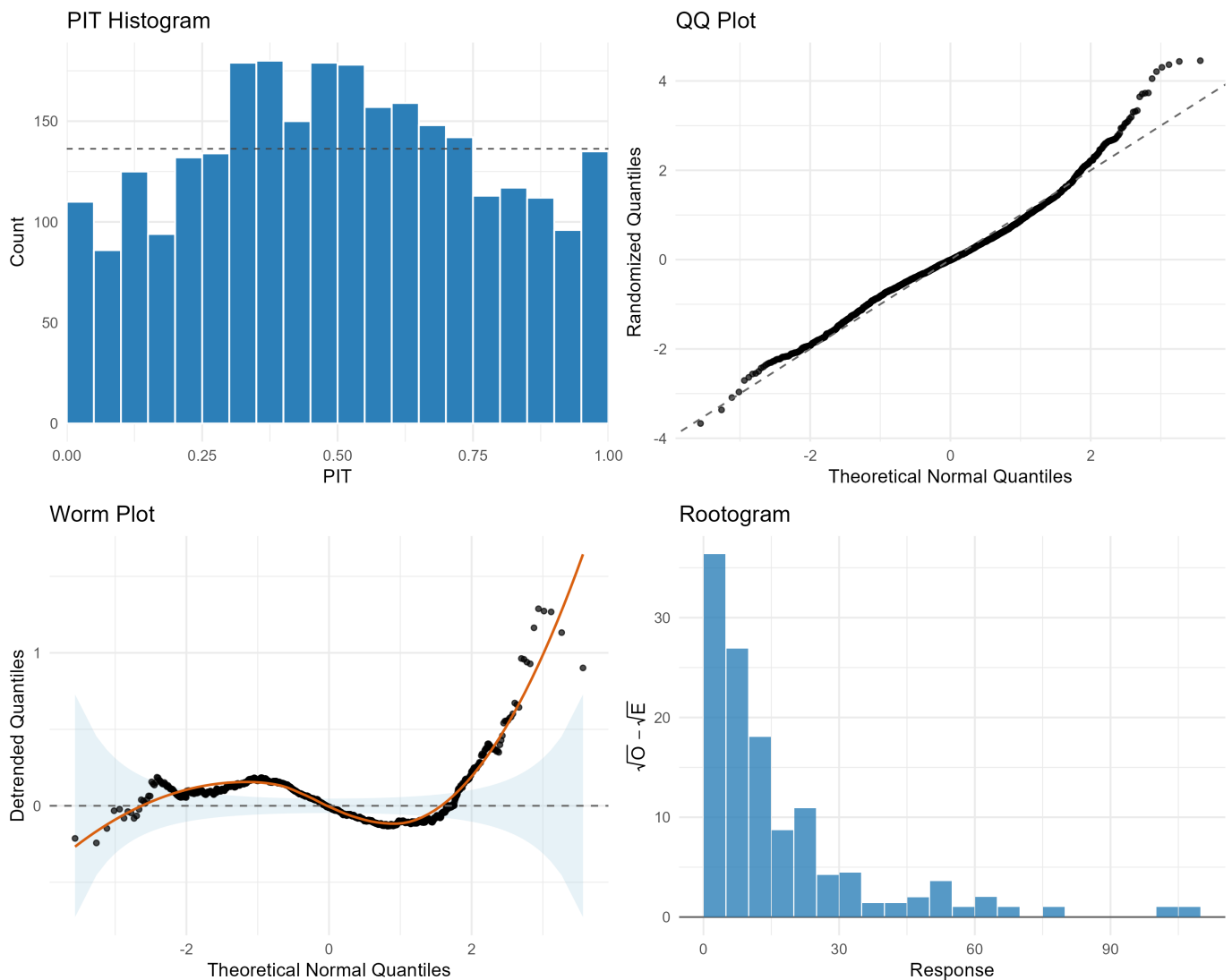


Term-level visual review for the final model.

Step 11 Plot: Marginal Diagnostics

The marginal diagnostic dashboard explains the `check_model` marginal failure. The PIT histogram is not flat and the QQ/worm plots show tail departures, which should be acknowledged in the paper rather than treated as a perfect application fit.

```
plot(final_fit, data = sample_data, randomize = TRUE)
```

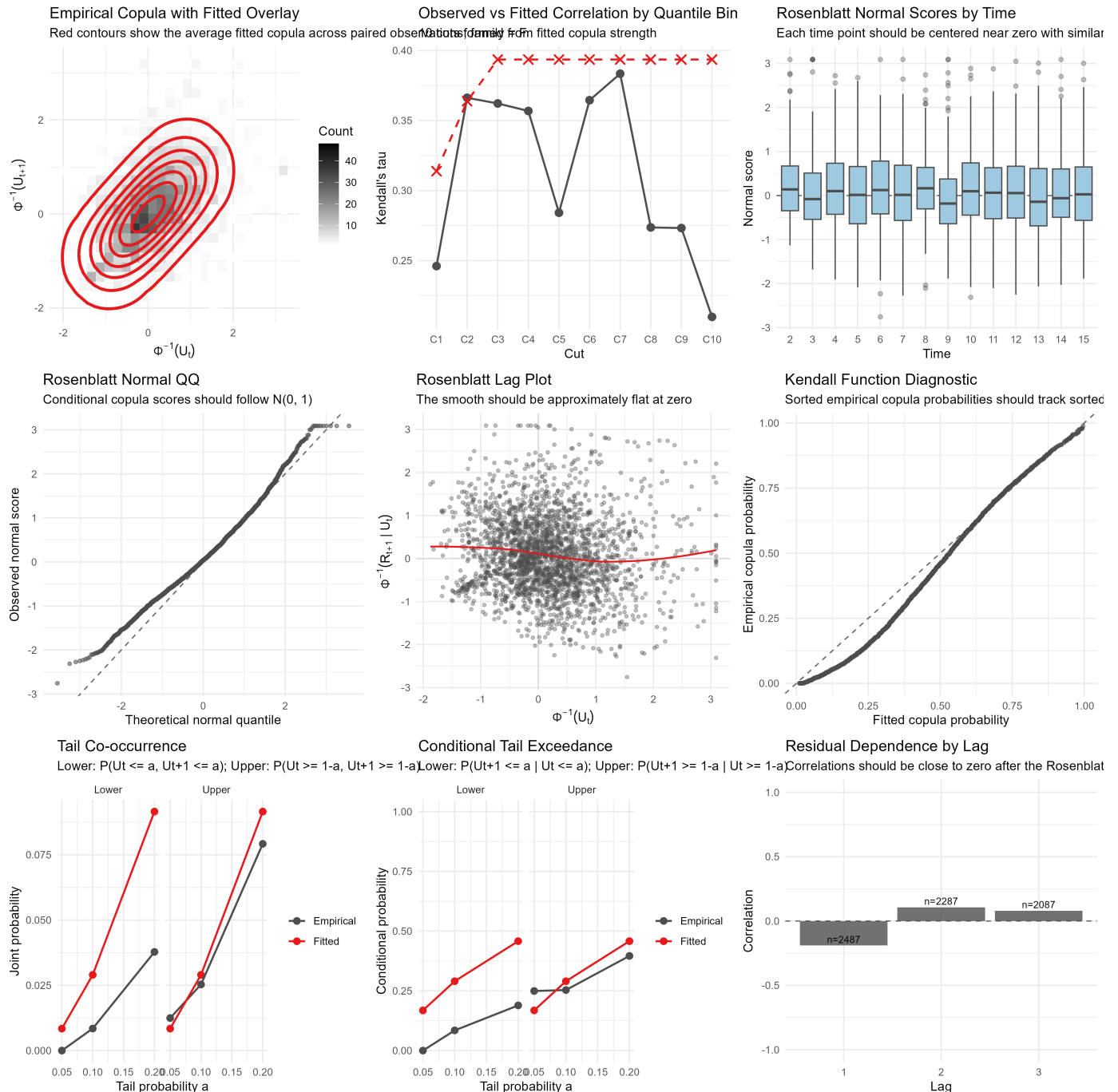


PIT, QQ, worm, and rootogram diagnostics from `plot(fit)`.

Step 12 Plot: Copula Diagnostics

The copula diagnostic dashboard is the strongest support for the application claim. It checks empirical copula shape, fitted versus observed correlation by quantile bin, Rosenblatt residuals, Kendall function calibration, tail co-occurrence, conditional tail exceedance, and residual dependence by lag.

```
diag <- plot_copula_diagnostics(
  final_fit,
  data = sample_data,
  plot = FALSE,
  dashboard_ncol = 3
)
diag$dashboard
```



Empirical copula, Rosenblatt residual, Kendall function, tail, and residual-lag diagnostics.

Copula Diagnostic Tables

The dashboard summaries are also written as CSVs so the paper can quote numerical diagnostics without re-parsing plots.

```
diag$quartile_summary  
diag$tail_summary  
diag$residual_lag_summary
```

lag	normal_score_cor
1.000	
2.000	
3.000	

Tail Co-occurrence Diagnostic

threshold	empirical_lower	fitted_lower	empirical_upper	fitted_upper
0.050				
0.050				
0.100				
0.100				
0.200				
0.200				

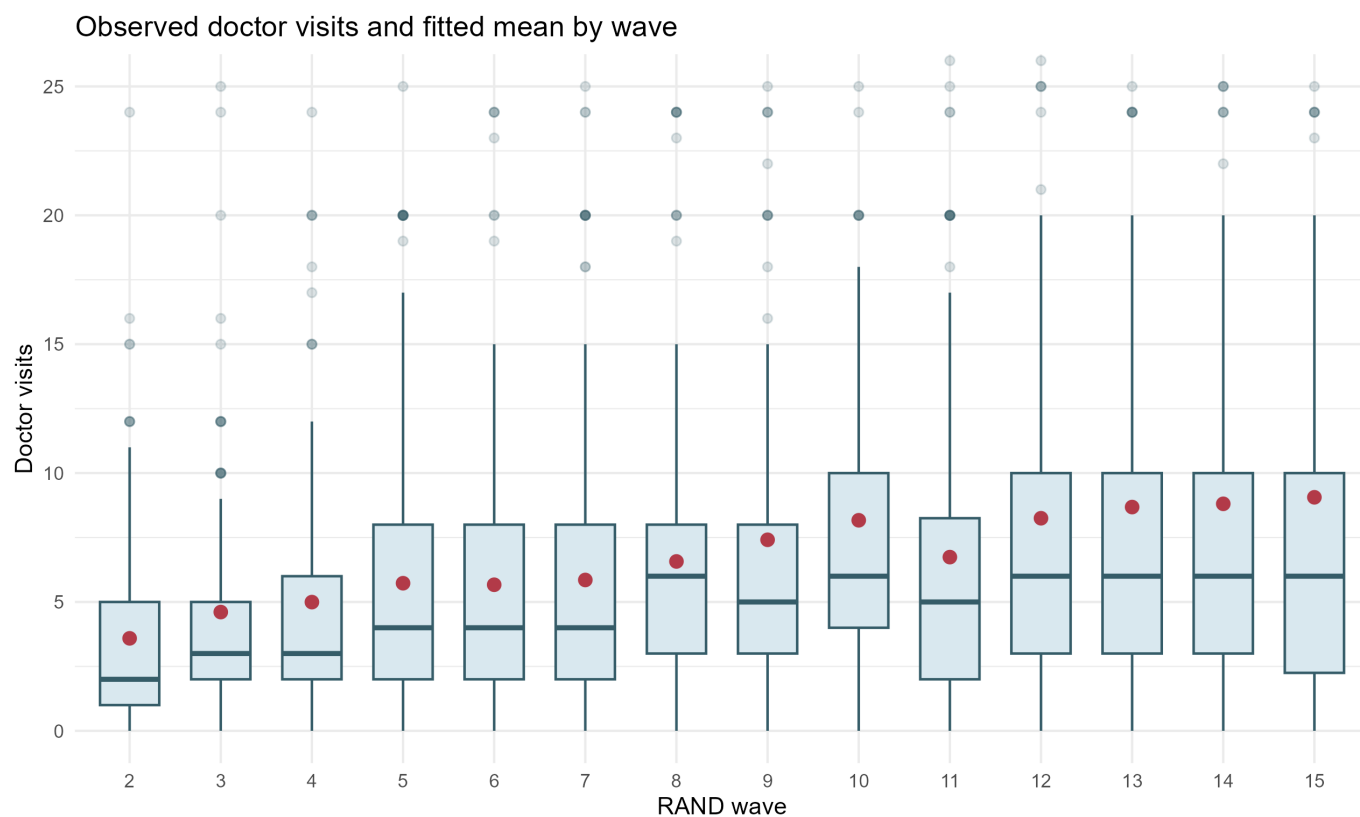
Conditional Tail Exceedance Diagnostic

threshold	empirical_lower	fitted_lower	empirical_upper	fitted_upper
0.050				
0.050				
0.100				
0.100				
0.200				
0.200				

Step 13 Plot: Observed Distribution and Fitted Mean

This paper-facing figure summarizes the response distribution over wave and overlays fitted means from the final model.

```
jss_rand_write_distribution_figure(sample_data, final_fit, paths$distribution_figure)
```

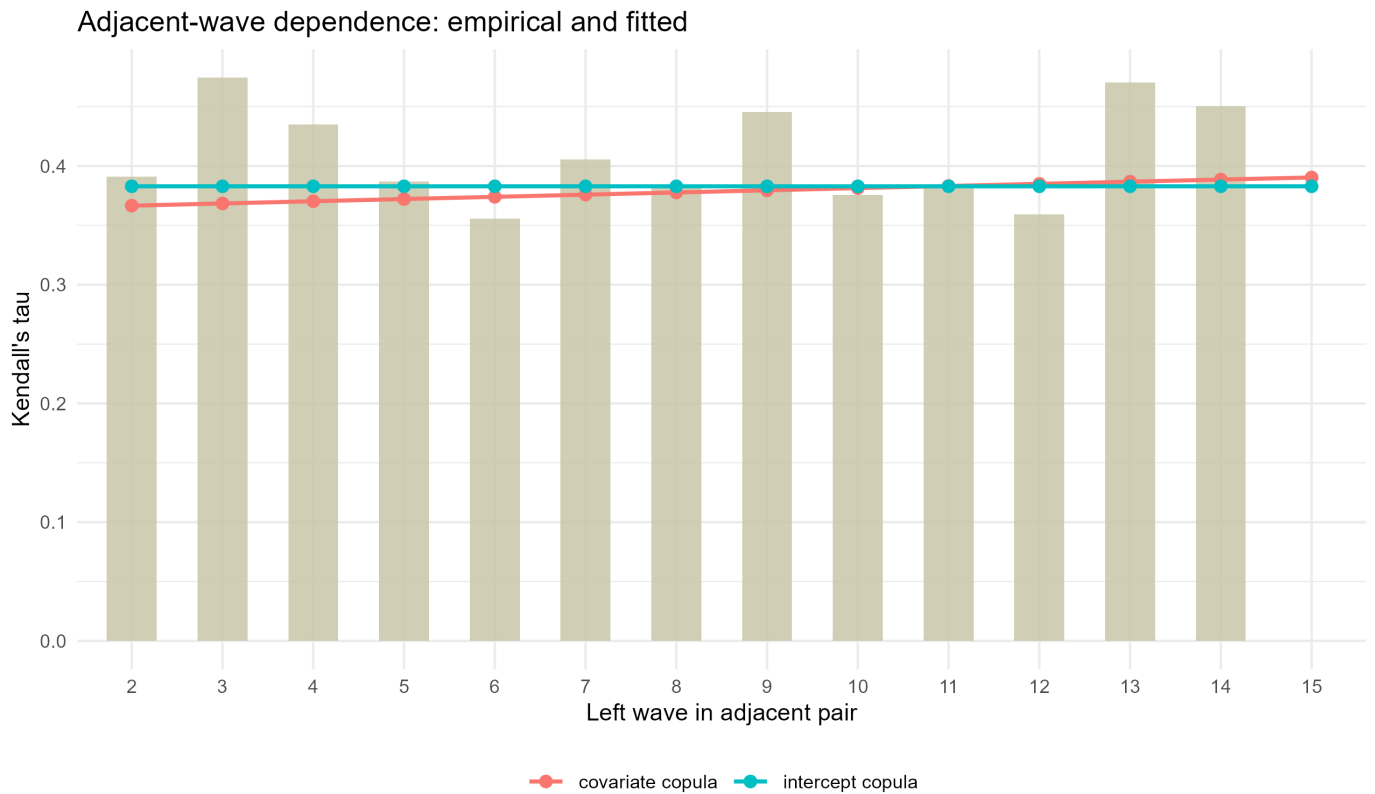


Observed doctor visits by wave with fitted mean overlay.

Step 14 Plot: Adjacent-Wave Dependence

This is the key longitudinal interpretation plot. It compares empirical adjacent-wave Kendall tau with fitted dependence under intercept-only and covariate-dependent theta models.

```
jss_rand_write_dependence_figure(  
  sample_data,  
  intercept_fit,  
  contrast_fit,  
  paths$figure  
)
```

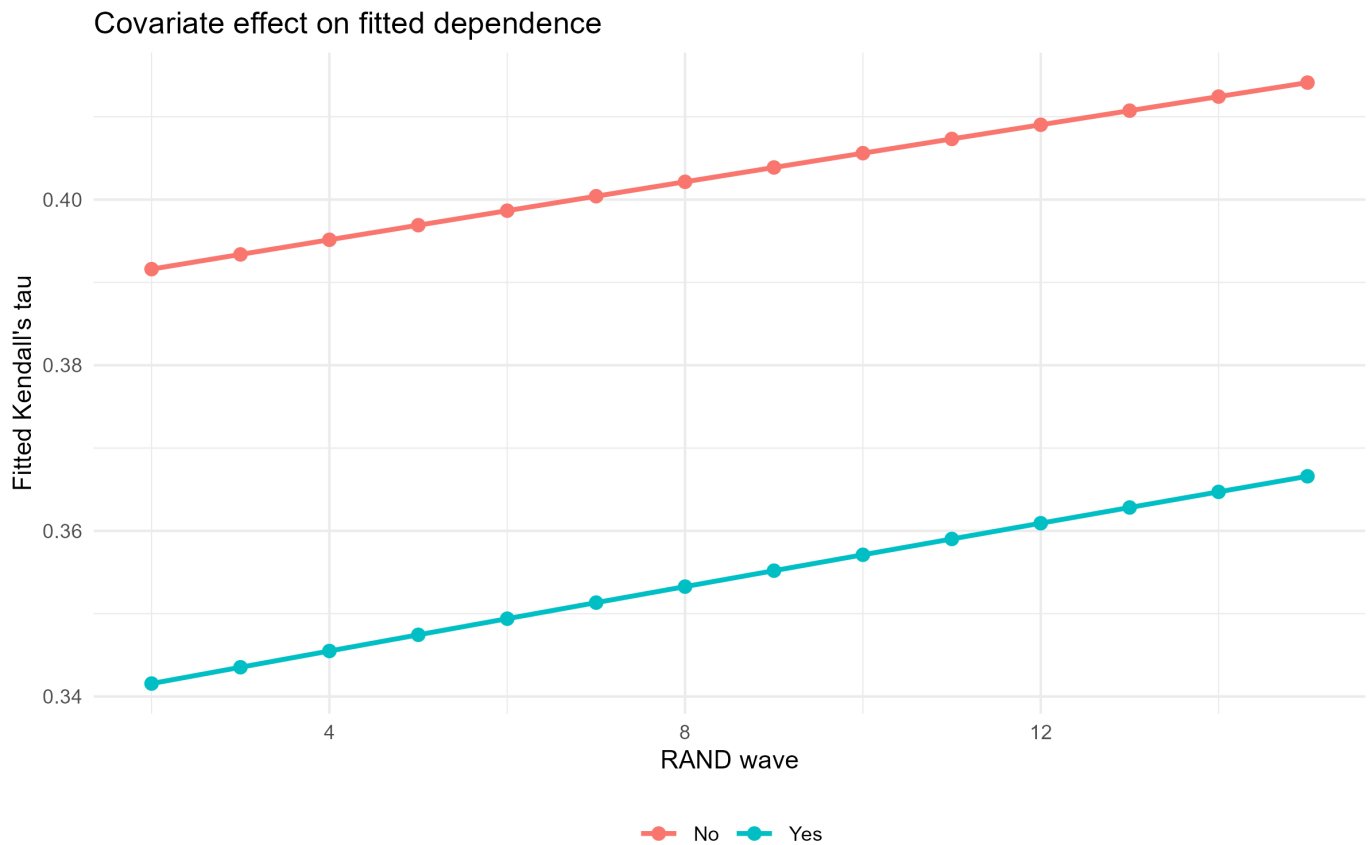


Empirical adjacent-wave Kendall tau and fitted copula dependence.

Step 15 Plot: Covariate Effect on Dependence

The theta effect plot translates the copula parameter into Kendall tau. This is the clearest visual for the paper claim: covariates can alter the longitudinal correlation structure after marginal effects are adjusted.

```
jss_rand_write_theta_effect_figure(  
  sample_data,  
  contrast_fit,  
  contrast$model,  
  paths$theta_figure  
)
```

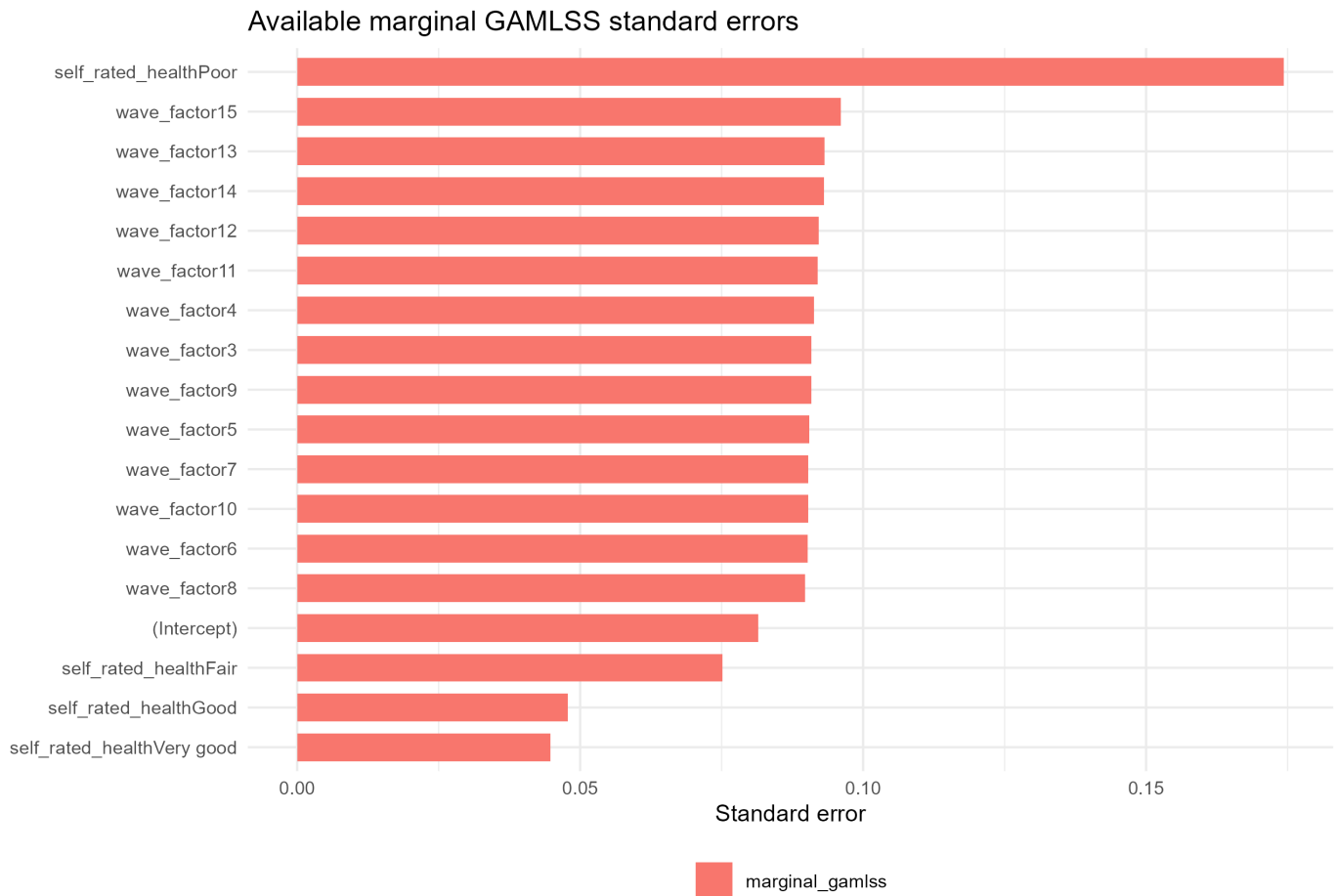


Covariate-dependence effect on fitted Kendall tau.

Step 16 Plot: Standard Error Comparison

The standard-error comparison records what is available for marginal GAMLSS and what is unavailable for exact-discrete longitudinal covariance calculations when the Hessian is singular or skipped. This guards against overclaiming precision.

```
se_comparison <- jss_rand_standard_error_comparison(  
  sample_data,  
  marginal_se_table,  
  intercept_fit,  
  contrast_fit  
)  
jss_rand_write_se_figure(se_comparison, paths$se_figure)
```



Available marginal standard errors and explicit unavailable longitudinal entries.

Step 17: Benchmark Standard Models

The native workflow includes benchmark comparisons against familiar longitudinal or mean-model baselines. These are directional rather than definitive, but they show whether the distributional copula model changes conclusions relative to standard mean-focused models.

```
bench <- benchmark_standard_models(
  data = observed,
  formula = doctor_visits ~ wave_factor + self_rated_health + hospital_stay +
    hypertension + cancer + education,
  subject_var = "subject",
  family = poisson(),
  comparators = c("glm", "gee", "glmm", "gam", "gamm"),
  fit = final_fit,
  fit_name = "gamlss_longitudinal"
)
```

comparator	estimator	available	requested	role
glm	stats::glm	TRUE	TRUE	independence mean baseline
gee	geepack::geeglm	TRUE	TRUE	marginal mean baseline with working correlation
glmm	lme4::lmer/glmer	TRUE	TRUE	random-intercept conditional mean baseline
gam	mgcv::gam	TRUE	TRUE	independence smooth mean baseline
gamm	mgcv::gam + s(subject, bs = 're')	TRUE	TRUE	smooth mean baseline with subject random-effect smooth
glmmTMB	glmmTMB::glmmTMB	FALSE	FALSE	optional flexible GLMM baseline

Benchmark Results

The model comparison table contains prediction and distributional metrics where available. Optional comparator failures or missing packages are retained explicitly.

method	success	rmse	mae	log_score	pit_ks_p_value	interval_coverage	error
gamlss_longitudinal	TRUE	7.245	4.248				
glm	TRUE	7.141	4.219				
gee	TRUE	7.139	4.242				
glmm	TRUE	7.384	4.158				
gam	TRUE	7.141	4.219				
gamm	FALSE						'names' attribute [1] must be the same length as the vector [0]

Step 18: Prediction Example

Prediction is performed on an aggregated reference wave grid rather than individual RAND rows, keeping the paper artifact private-safe while demonstrating the prediction API.

```
newdata <- jss_rand_reference_prediction_grid(sample_data)
pred <- predict(final_fit, newdata = newdata, type = "mean",
se.fit = TRUE, interval = "confidence")
```

wave	wave_scaled	survey_year	fit	se.fit	lower	upper
2.000	0.00e+00	1994.0	3.427	0.00e+00		
3.000	0.077	1996.0	4.251	0.268		
4.000	0.154	1998.0	4.221	0.303		
5.000	0.231	2000.0	4.739	0.359		
6.000	0.308	2002.0	4.658	0.348		
7.000	0.385	2004.0	4.578	0.329		
8.000	0.462	2006.0	4.915	0.297		
9.000	0.538	2008.0	5.146	0.315		
10.00	0.615	2010.0	5.452	0.382		
11.00	0.692	2012.0	4.352	0.327		
12.00	0.769	2014.0	5.171	0.389		
13.00	0.846	2016.0	5.186	0.388		
14.00	0.923	2018.0	5.020	0.386		
15.00	1.000	2020.0	4.613	0.380		

Step 19: Marginal Effects Example

The marginal-effects example uses the standard package helper to summarize a counterfactual covariate contrast. Here it reports the response-scale mu contrast for hospital stay.

```
effects <- marginal_effects(
  final_fit,
  newdata = observed,
  variable = "hospital_stay",
  values = levels(sample_data$hospital_stay),
  parameter = "mu",
  se.fit = FALSE
)
```

variable	value	parameter	estimate	reference	contrast
hospital_stay	No	mu	6.005	No	0.00e+00
hospital_stay	Yes	mu	9.072	No	3.067

Step 20: Simulation Summary

The final native workflow tool is simulation from the fitted copula model. The table and plot compare observed and simulated wave means, giving a simple posterior-predictive-style check of trajectory-level behaviour.

```
sim_model <- simulate(final_fit, nsim = 25, seed = 20260528)
simulation_summary <- summarize_by_wave(sample_data, sim_model)
```

wave	observed_mean	observed_variance	simulated_mean	simulated_variance
2.000	3.702	28.24	3.568	15.97
3.000	4.680	43.64	4.549	21.15
4.000	5.427	67.38	4.866	28.43
5.000	5.600	40.83	5.863	41.53
6.000	5.808	38.66	5.570	33.82
7.000	5.910	41.95	5.846	38.64
8.000	7.060	76.88	6.588	50.42
9.000	7.143	73.77	7.285	65.32
10.00	8.333	98.71	8.159	79.01
11.00	6.235	30.43	6.529	57.18

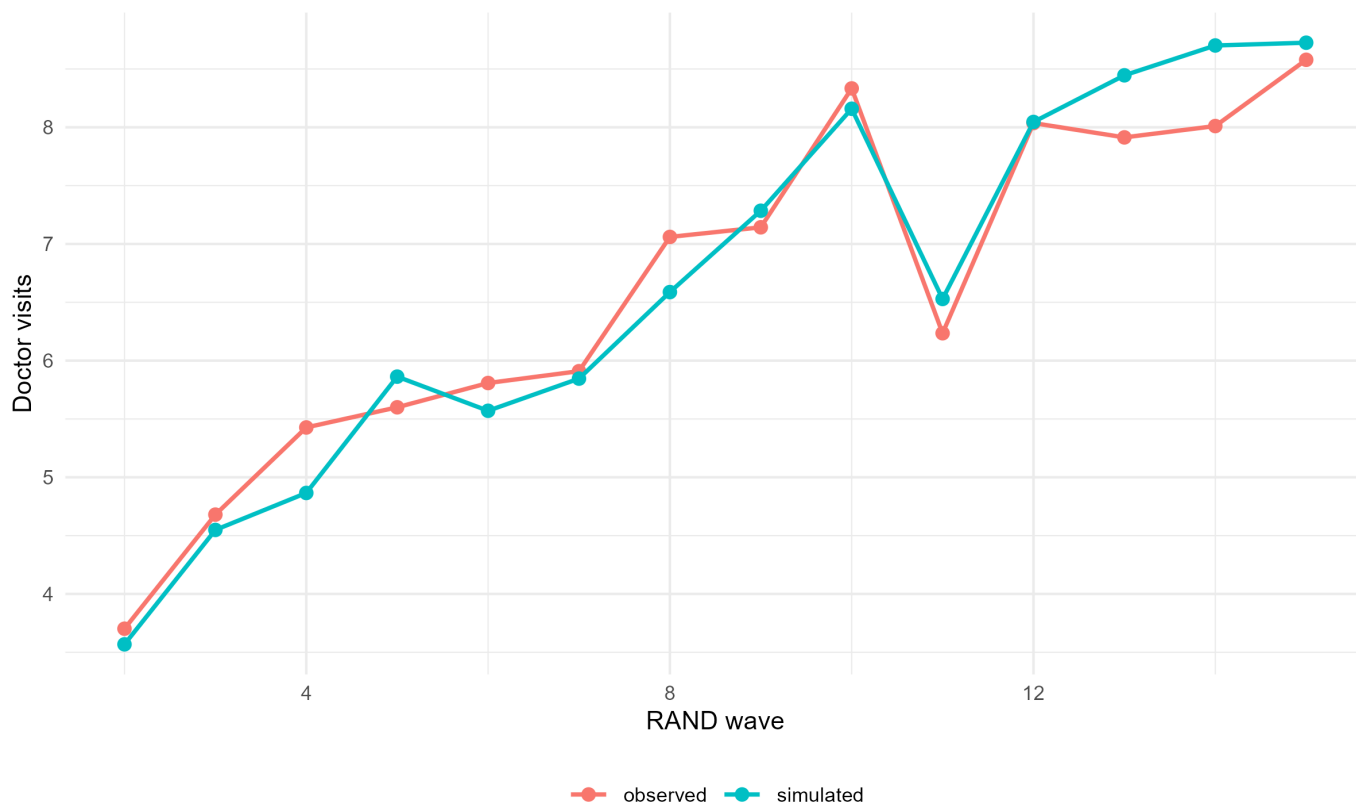
wave	observed_mean	observed_variance	simulated_mean	simulated_variance
12.00	8.036	70.49	8.046	72.12
13.00	7.914	68.78	8.446	85.59
14.00	8.011	64.81	8.701	101.23
15.00	8.579	76.70	8.725	106.07

Step 20 Plot: Simulation Check

Model-based simulations preserve the fitted copula dependence and marginal structure. The plotted wave means show that simulated trajectories broadly reproduce the observed mean pattern.

```
simulate(final_fit, nsim = 25, seed = 20260528)
```

Observed and simulated mean doctor visits



Observed and model-simulated mean doctor visits by wave.

Interpretive Bottom Line

The expanded RAND workflow supports a covariate-dependent dependence structure: the AIC-preferred theta model uses hospital stay under a Frank copula, and the pre-specified wave-plus-arthritis contrast still improves likelihood relative to the intercept-only dependence model. The copula diagnostics indicate that modelling dependence materially reduces residual longitudinal association, while the marginal diagnostics warn that NBI calibration is not perfect in the full PIT sense. For the paper, the cleanest claim is therefore methodological and diagnostic: allowing theta to vary by health covariates changes the fitted longitudinal association and should be considered when interpreting marginal effects and uncertainty, but the application should transparently report remaining marginal calibration limitations.

```
# Complete module entrypoint
settings <- jss_settings()
out <- jss_run_06_rand_doctor_visits(settings)

# Key generated paper outputs
out$tables
out$figures
out$notes
```